



AI - Machine Learning Engineer

QP Code: SSC/Q8113

Version: 2.0

NSQF Level: 7

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SSC/Q8113: AI - Machine Learning Engineer

Brief Job Description

Individuals at this job are responsible for developing applications and platforms in AI & Big Data Analytics. S/he will be responsible for developing software code to deploy algorithmic models as per the needs of the business and evaluating the technical performance of the same

Personal Attributes

A Machine Learning engineer needs to have strong analytical and logical skills, attention to detail and problem solving ability. S/he needs to have good communication skills to work with various stakeholders that includes Data Scientists, Data Architects, Business Analysts etc. to ensure alignment on business objectives

Applicable National Occupational Standards (NOS)

Compulsory NOS:

1. [SSC/N8121: Evaluate technical performance of algorithmic models](#)
2. [SSC/N8122: Develop software code to support the deployment of algorithmic models](#)
3. [SSC/N9001: Manage your work to meet requirements](#)
4. [SSC/N9002: Work effectively with colleagues](#)
5. [SSC/N9004: Provide data/information in standard formats](#)
6. [SSC/N9006: Build and Maintain relationships in a Workplace](#)
7. [SSC/N9014: Maintain an inclusive, environmentally sustainable workplace](#)

Qualification Pack (QP) Parameters

Sector	IT-ITes
Sub-Sector	Future Skills
Occupation	Artificial Intelligence and Big Data Analytics
Country	India
NSQF Level	7



Aligned to NCO/ISCO/ISIC Code	NCO-2015/2512.NIL
Minimum Educational Qualification & Experience	Graduate (Engineering/Technology/Statistics/Mathematics/Computer Science) with 2-3 Years of experience Recommended
Minimum Level of Education for Training in School	
Pre-Requisite License or Training	NA
Minimum Job Entry Age	21 Years
Last Reviewed On	14/04/2020
Next Review Date	14/04/2025
Deactivation Date	14/04/2025
NSQC Approval Date	
Version	2.0

SSC/N8121: Evaluate technical performance of algorithmic models

Description

This unit is about measuring the technical performance of algorithmic models for deployment.

Scope

This unit/task covers the following:

- Define model parameters
- Analyse technical constraints
- Technical Performance Constraints: speed, runtime, memory

Elements and Performance Criteria

Define model parameters

To be competent, the user/individual on the job must be able to:

PC1. define technical specifications and limitations of the system running the algorithmic model

PC2. define the data flows and evaluate the data structures in the algorithmic model

Analyse technical constraints

To be competent, the user/individual on the job must be able to:

PC3. evaluate the speed, runtime and memory interdependencies of the algorithmic model

PC4. evaluate the parallel programming constraints of the system

PC5. modify and test the algorithmic model to fit speed, runtime and memory limitations of the system

PC6. update documentation on the algorithmic model

PC7. provide feedback on model performance to the team responsible for designing and developing the model

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

KU1. the purpose and aims of the system being developed

KU2. organizational policies, procedures and guidelines for evaluating the technical performance of algorithmic models

KU3. different data sources and how to access the required documents and information from data sources

KU4. organizational policies and procedures for sharing data

KU5. organizational policies and procedures for documenting algorithmic models

KU6. who to involve when evaluating the technical performance of algorithmic models

KU7. the range of standard templates and tools available and how to use them

KU8. how to identify and interpret the technical specifications and limitations of a system

KU9. different algorithmic models

- KU10.** different data flows and data structures
- KU11.** how to evaluate the speed, runtime and memory dependencies of algorithmic models
- KU12.** different parallel computing systems such as SISD (Single Instruction Single Data Stream), SIMD (Single Instruction Multiple Data Streams), MISD (Multiple Instructions Single Data Stream), MIMD (Multiple Instructions Multiple Data Streams)
- KU13.** different methodological approaches to measure algorithmic model performance such as Big O notation
- KU14.** how to identify and refer anomalies
- KU15.** how to work on various operating systems such as linux, ubuntu, or windows

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** make decisions on suitable courses of action.
- GS2.** plan and organize your own work to achieve targets and deadlines
- GS3.** refer anomalies to the supervisor
- GS4.** check your work is complete and free from errors

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Define model parameters</i>	10	20	-	-
PC1. define technical specifications and limitations of the system running the algorithmic model	5	10	-	-
PC2. define the data flows and evaluate the data structures in the algorithmic model	5	10	-	-
<i>Analyse technical constraints</i>	20	50	-	-
PC3. evaluate the speed, runtime and memory interdependencies of the algorithmic model	5	10	-	-
PC4. evaluate the parallel programming constraints of the system	5	10	-	-
PC5. modify and test the algorithmic model to fit speed, runtime and memory limitations of the system	5	15	-	-
PC6. update documentation on the algorithmic model	-	5	-	-
PC7. provide feedback on model performance to the team responsible for designing and developing the model	5	10	-	-
NOS Total	30	70	-	-

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8121
NOS Name	Evaluate technical performance of algorithmic models
Sector	IT-ITes
Sub-Sector	Future Skills
Occupation	Artificial Intelligence & Big Data Analytics
NSQF Level	7
Credits	TBD
Version	1.0
Last Reviewed Date	17/10/2018
Next Review Date	31/12/2019
NSQC Clearance Date	19/12/2018

SSC/N8122: Develop software code to support the deployment of algorithmic models

Description

This unit is about developing software code to support the deployment of algorithmic models to meet specific requirements and constraints.

Scope

This unit/task covers the following:

- Define requirements
- Convert specifications into code
- Perform unit tests
- Software Requirements Specification: functional requirements, non-functional requirements
- High Level Design: technical solution selected and its rationale, assumptions, constraints, dependencies, interfaces
- Low Level Design: programming structures, coding standards, checklists

Elements and Performance Criteria

Define requirements

To be competent, the user/individual on the job must be able to:

- PC1.** define technical specifications and limitations of the autonomous system
- PC2.** evaluate the design of data flows and structures in the autonomous system
- PC3.** evaluate the design of core algorithmic models in the autonomous system
- PC4.** evaluate the parallel programming constraints of the autonomous system

Convert specifications into code

To be competent, the user/individual on the job must be able to:

- PC5.** integrate technical specifications, data flows, core algorithmic models and parallel programming constraints and convert them into reusable code
- PC6.** create documentation on developed software code for relevant stakeholders
- PC7.** validate developed code with relevant stakeholders

Perform unit tests

To be competent, the user/individual on the job must be able to:

- PC8.** create and execute unit test cases to analyse the performance of the code
- PC9.** evaluate the defects in the code and identify corrective actions
- PC10.** implement corrective actions to optimize code performance
- PC11.** submit tested and optimized code for approval by appropriate people

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the purpose and aims of the autonomous system being developed
- KU2.** organizational policies, procedures and guidelines for developing software code
- KU3.** different data sources and how to access the required documents and information from data sources
- KU4.** organizational policies and procedures for sharing data
- KU5.** organizational policies and procedures for documenting developed code
- KU6.** who to involve when developing software code
- KU7.** their organization's approval process for software code designs
- KU8.** the range of standard templates and tools available and how to use them
- KU9.** how to identify and interpret the technical specifications and limitations of an autonomous system
- KU10.** different data flows and data structures
- KU11.** different algorithmic models and their design
- KU12.** different parallel computing systems such as SISD (Single Instruction Single Data Stream), SIMD (Single Instruction Multiple Data Streams), MISD (Multiple Instructions Single Data Stream), MIMD (Multiple Instructions Multiple Data Streams)
- KU13.** the range of code generation tools and unit testing tools used to develop software code
- KU14.** how to use coding tools
- KU15.** how to create, review and execute unit test cases
- KU16.** how to determine whether components are suitable for re-use
- KU17.** different types of problems and defects that may occur and how these may be resolved
- KU18.** how recording corrective actions for problems and defects can improve future designs
- KU19.** how to use corrective actions to optimize performance of software code
- KU20.** how to identify and refer anomalies
- KU21.** how to work on various operating systems such as linux, ubuntu, or windows

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** make a decision on a suitable course of action
- GS2.** apply problem-solving approaches in different situations
- GS3.** analyze data and activities
- GS4.** configure data and disseminate relevant information to others
- GS5.** apply balanced judgments to different situations
- GS6.** check your work is complete and free from errors

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Define requirements</i>	11	24	-	-
PC1. define technical specifications and limitations of the autonomous system	2	3	-	-
PC2. evaluate the design of data flows and structures in the autonomous system	3	7	-	-
PC3. evaluate the design of core algorithmic models in the autonomous system	3	7	-	-
PC4. evaluate the parallel programming constraints of the autonomous system	3	7	-	-
<i>Convert specifications into code</i>	5	25	-	-
PC5. integrate technical specifications, data flows, core algorithmic models and parallel programming constraints and convert them into reusable code	5	15	-	-
PC6. create documentation on developed software code for relevant stakeholders	-	5	-	-
PC7. validate developed code with relevant stakeholders	-	5	-	-
<i>Perform unit tests</i>	9	26	-	-
PC8. create and execute unit test cases to analyse the performance of the code	3	7	-	-
PC9. evaluate the defects in the code and identify corrective actions	3	7	-	-
PC10. implement corrective actions to optimize code performance	3	7	-	-
PC11. submit tested and optimized code for approval by appropriate people	-	5	-	-
NOS Total	25	75	-	-

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N8122
NOS Name	Develop software code to support the deployment of algorithmic models
Sector	IT-ITes
Sub-Sector	Future Skills
Occupation	Artificial Intelligence & Big Data Analytics
NSQF Level	7
Credits	TBD
Version	1.0
Last Reviewed Date	17/10/2018
Next Review Date	31/12/2019
NSQC Clearance Date	19/12/2018

SSC/N9001: Manage your work to meet requirements

Description

This unit is about planning and organizing your work in order to complete it to the required standards on time.

Scope

The scope covers the following :

- Utilise resources
- Ensure compliance

Elements and Performance Criteria

To be competent, the user/individual on the job must be able to:

- PC1.** establish and agree your work requirements with appropriate people
- PC2.** keep the immediate work area clean and tidy
- PC3.** utilize time effectively
- PC4.** use resources correctly and efficiently
- PC5.** treat confidential information correctly
- PC6.** work in line with the organization's policies and procedures
- PC7.** work within the limits of the job role
- PC8.** obtain guidance from appropriate people, where necessary
- PC9.** ensure the work meets the agreed requirements

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the priorities for the area of work
- KU2.** role, responsibilities, limits of the responsibilities and whom these must be agreed with, as well as when to involve others
- KU3.** the importance of having a tidy work area and how to do this
- KU4.** how to prioritize your workload according to urgency and importance and the benefits of this
- KU5.** the organizations policies and procedures, especially for dealing with confidential information, and the importance of complying with these
- KU6.** the purpose of keeping others updated with the progress of the work
- KU7.** the purpose and value of being flexible and adapting work plans to reflect change
- KU8.** the importance of completing work accurately and how to do this
- KU9.** appropriate timescales for completing the work and the implications of not meeting these for self and the organization
- KU10.** resources needed for the work and how to obtain and use these

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read instructions, guidelines, procedures, rules and service level agreements
- GS2.** ask for clarification and advice from line managers
- GS3.** communicate orally with colleagues
- GS4.** make decisions on suitable courses
- GS5.** plan and organize the work to achieve targets and deadlines
- GS6.** agree to objectives and work requirements
- GS7.** deliver consistent and reliable service to customers
- GS8.** check that the work meets customer requirements
- GS9.** refer anomalies to the line manager
- GS10.** seek clarification on problems from others
- GS11.** provide relevant information to others
- GS12.** analyze needs, requirements and dependencies in order to meet the work requirements
- GS13.** apply judgments to different situations
- GS14.** ensure the work is complete and free from errors
- GS15.** get the work checked by peers
- GS16.** work effectively in a team environment
- GS17.** use information technology effectively, to input and/or extract data accurately
- GS18.** identify and refer anomalies in data
- GS19.** store and retrieve information
- GS20.** keep up to date with changes, procedures and practices in the role

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
	25	75	-	-
PC1. establish and agree your work requirements with appropriate people	-	6.25	-	-
PC2. keep the immediate work area clean and tidy	6.25	6.25	-	-
PC3. utilize time effectively	6.25	6.25	-	-
PC4. use resources correctly and efficiently	6.25	12.5	-	-
PC5. treat confidential information correctly	-	6.25	-	-
PC6. work in line with the organization's policies and procedures	-	12.5	-	-
PC7. work within the limits of the job role	-	6.25	-	-
PC8. obtain guidance from appropriate people, where necessary	-	6.25	-	-
PC9. ensure the work meets the agreed requirements	6.25	12.5	-	-
NOS Total	25	75	-	-

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N9001
NOS Name	Manage your work to meet requirements
Sector	IT-ITeS
Sub-Sector	IT Services, Business Process Management, Engineering R&D, Software Product Development, IT Support Services, Software Products, Future Skills
Occupation	Generic
NSQF Level	4
Credits	TBD
Version	2.0
Last Reviewed Date	NA
Next Review Date	31/03/2025
NSQC Clearance Date	

SSC/N9002: Work effectively with colleagues

Description

This unit is about working effectively with colleagues, either in your own work group or in other work groups within your organization.

Scope

The scope covers the following :

- Communicate with colleagues
- Show respect

Elements and Performance Criteria

Communicate with colleagues

To be competent, the user/individual on the job must be able to:

- PC1.** communicate with colleagues clearly, concisely and accurately
- PC2.** work with colleagues to integrate the work effectively with theirs
- PC3.** pass on essential information to colleagues in line with organizational requirements

Show respect

To be competent, the user/individual on the job must be able to:

- PC4.** work in ways that show respect for colleagues
- PC5.** carry out commitments one has made to colleagues
- PC6.** identify any problems while working with colleagues and take the initiative to solve these problems
- PC7.** follow the organization's policies and procedures for working with colleagues

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the organization's policies and procedures for working with colleagues and the role and responsibilities in relation to this
- KU2.** the importance of effective communication and establishing good working relationships with colleagues
- KU3.** different methods of communication and the circumstances in which it is appropriate to use these
- KU4.** benefits of developing productive working relationships with colleagues
- KU5.** the importance of creating an environment of trust and mutual respect in an environment where there is no authority over those working with
- KU6.** where you do not meet the commitments, the implications this will have on individuals and the organization
- KU7.** different types of information that colleagues might need and the importance of providing this information when it is required

KU8. the importance of understanding problems from the colleagues perspective and how to provide support, where necessary, to resolve these

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete accurate, well written work with attention to detail
- GS2.** communicate effectively with colleagues in writing
- GS3.** read instructions, guidelines, procedures, rules and service level agreements
- GS4.** make decisions on suitable courses
- GS5.** ask for clarification and advice from line managers
- GS6.** help reach agreements with colleagues
- GS7.** plan and organize the work to achieve targets and deadlines
- GS8.** ensure the work meets customer requirements, and deliver consistent and reliable service
- GS9.** apply problem solving approaches in different situations
- GS10.** apply balanced judgments to different situations
- GS11.** ensure the work is complete and free from errors
- GS12.** ensure the work is complete and free from errors
- GS13.** work effectively with colleagues and other teams in a team environment
- GS14.** treat other cultures with respect
- GS15.** identify and refer anomalies
- GS16.** keep up to date with changes, procedures and practices in the role

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Communicate with colleagues</i>	18	30	-	-
PC1. communicate with colleagues clearly, concisely and accurately	-	20	-	-
PC2. work with colleagues to integrate the work effectively with theirs	-	10	-	-
PC3. pass on essential information to colleagues in line with organizational requirements	18	-	-	-
<i>Show respect</i>	2	50	-	-
PC4. work in ways that show respect for colleagues	2	20	-	-
PC5. carry out commitments one has made to colleagues	-	10	-	-
PC6. identify any problems while working with colleagues and take the initiative to solve these problems	-	10	-	-
PC7. follow the organization's policies and procedures for working with colleagues	-	10	-	-
NOS Total	20	80	-	-

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N9002
NOS Name	Work effectively with colleagues
Sector	IT-ITeS
Sub-Sector	IT Services, Business Process Management, Engineering R&D, Software Product Development, IT Support Services, Software Products, Future Skills
Occupation	Generic
NSQF Level	4
Credits	TBD
Version	2.0
Last Reviewed Date	NA
Next Review Date	31/03/2025
NSQC Clearance Date	

SSC/N9004: Provide data/information in standard formats

Description

This unit is about providing specified data/information related to your work in templates or other standard formats.

Scope

The scope covers the following :

- Obtain information
- Analyze and report information

Elements and Performance Criteria

Obtain information

To be competent, the user/individual on the job must be able to:

- PC1.** establish and agree with appropriate people the data/information you need to provide, the formats in which you need to provide it, and when you need to provide it
- PC2.** obtain the data/information from reliable sources
- PC3.** check that the obtained data/information is accurate, complete and up-to-date
- PC4.** obtain advice or guidance from appropriate people where there are problems with the data/information

Analyze and report information

To be competent, the user/individual on the job must be able to:

- PC5.** carry out rule-based analysis of the data/information, if required
- PC6.** insert the data/information into the agreed formats
- PC7.** report any unresolved anomalies in the data/ information to appropriate people
- PC8.** provide complete, accurate and up-to-date data/information to the appropriate people in the required formats on time

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the organization's procedures and guidelines for providing data/information in standard formats and the role and responsibilities in relation to this
- KU2.** the knowledge management culture of the organization
- KU3.** the organization's policies and procedures for recording and sharing information and the importance of complying with these
- KU4.** the importance of validating data/information before use and how to do this
- KU5.** procedures for updating data in appropriate formats and with proper validation
- KU6.** the purpose of the CRM database
- KU7.** how to use the CRM database to record and extract information

- KU8.** the importance of having data/information reviewed by others
- KU9.** the scope of any data/information requirements including the level of detail required
- KU10.** the importance of keeping within the scope of work and adhering to timescales
- KU11.** data/information one may need to provide including the sources and how to do this
- KU12.** templates and formats used for data/information including their purpose and how to use these
- KU13.** different techniques used to obtain data/information and how to apply these
- KU14.** rule-based analysis on the data/information
- KU15.** typical anomalies that may occur in data/information
- KU16.** whom to go to in the event of inaccurate data/information and how to report this

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** complete accurate, well written work with attention to detail
- GS2.** read instructions, guidelines, procedures, rules and service level agreements
- GS3.** listen effectively and orally communicate information accurately
- GS4.** follow rule-based decision-making processes
- GS5.** make decisions on suitable courses of action
- GS6.** plan and organize the work to achieve targets and deadlines
- GS7.** check the work meets customer requirements and exceed customer expectations
- GS8.** apply problem solving approaches in different situations
- GS9.** configure data and disseminate relevant information to others
- GS10.** apply balanced judgments to different situations
- GS11.** use information technology effectively, to input and/or extract data accurately
- GS12.** validate and update data
- GS13.** store and retrieve information

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Obtain information</i>	18.75	31.25	-	-
PC1. establish and agree with appropriate people the data/information you need to provide, the formats in which you need to provide it, and when you need to provide it	12.5	-	-	-
PC2. obtain the data/information from reliable sources	-	12.5	-	-
PC3. check that the obtained data/information is accurate, complete and up-to-date	6.25	6.25	-	-
PC4. obtain advice or guidance from appropriate people where there are problems with the data/information	-	12.5	-	-
<i>Analyze and report information</i>	6.25	43.75	-	-
PC5. carry out rule-based analysis of the data/information, if required	-	25	-	-
PC6. insert the data/information into the agreed formats	-	12.5	-	-
PC7. report any unresolved anomalies in the data/ information to appropriate people	6.25	-	-	-
PC8. provide complete, accurate and up-to-date data/information to the appropriate people in the required formats on time	-	6.25	-	-
NOS Total	25	75	-	-

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N9004
NOS Name	Provide data/information in standard formats
Sector	IT-ITeS
Sub-Sector	IT Services, Business Process Management, Engineering R&D, Software Product Development, IT Support Services, Software Products, Future Skills
Occupation	Generic
NSQF Level	4
Credits	TBD
Version	2.0
Last Reviewed Date	NA
Next Review Date	31/03/2025
NSQC Clearance Date	

SSC/N9006: Build and Maintain relationships in a Workplace

Description

This unit is about building and maintaining constructive relationships at the workplace.

Scope

The scope covers the following :

- This unit/task covers the following: Build relationships, Maintain relationships
- Appropriate people: line manager, members of the team/department, members from other teams/department

Elements and Performance Criteria

Build relationships

To be competent, the user/individual on the job must be able to:

- PC1.** build rapport with appropriate people at the workplace
- PC2.** develop new professional relationships
- PC3.** build alliances to establish mutually beneficial working arrangements
- PC4.** foster an environment where others feel respected
- PC5.** identify and engage a diverse range of influential contacts

Maintain relationships

To be competent, the user/individual on the job must be able to:

- PC6.** obtain guidance from appropriate people, where necessary
- PC7.** attentively listen to ideas and give constructive feedback
- PC8.** promptly resolve conflicts between team members
- PC9.** work with colleagues to deliver shared goals
- PC10.** recognize the contributions made by your colleagues

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** organizational policies and procedures for building relationships and their role and responsibilities in relation to this
- KU2.** training programs to enable the development of relevant behavioural competencies
- KU3.** the importance of creating an environment of trust and mutual respect in the organisation
- KU4.** the importance of effective communication in developing productive working relationships with colleagues
- KU5.** training programs to enable the development of relevant behavioural competencies

Generic Skills (GS)



User/individual on the job needs to know how to:

GS1. ask for clarification and advice from line managers

GS2. work effectively in a team environment

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Build relationships</i>	17	33	-	-
PC1. build rapport with appropriate people at the workplace	3	7	-	-
PC2. develop new professional relationships	3	7	-	-
PC3. build alliances to establish mutually beneficial working arrangements	3	7	-	-
PC4. foster an environment where others feel respected	4	6	-	-
PC5. identify and engage a diverse range of influential contacts	4	6	-	-
<i>Maintain relationships</i>	13	37	-	-
PC6. obtain guidance from appropriate people, where necessary	3	7	-	-
PC7. attentively listen to ideas and give constructive feedback	3	7	-	-
PC8. promptly resolve conflicts between team members	2	8	-	-
PC9. work with colleagues to deliver shared goals	2	8	-	-
PC10. recognize the contributions made by your colleagues	3	7	-	-
NOS Total	30	70	-	-

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N9006
NOS Name	Build and Maintain relationships in a Workplace
Sector	IT-ITes
Sub-Sector	Future Skills
Occupation	Artificial Intelligence & Big Data Analytics
NSQF Level	6
Credits	TBD
Version	3.0
Last Reviewed Date	NA
Next Review Date	31/03/2025
NSQC Clearance Date	

SSC/N9014: Maintain an inclusive, environmentally sustainable workplace

Description

The unit is about implementing and improving diversity equality and inclusion in a sustainable and environment friendly workplace.

Scope

The scope covers the following :

- The scope covers the following: Enrich policies to respect diversity, Reinforce practices/regulations/policies to promote and improve equity (equality)/inclusivity, Emphasize sustainable environmental practices

Elements and Performance Criteria

Sustainable Practices

To be competent, the user/individual on the job must be able to:

- PC1.** optimize usage of electricity/energy, materials, and water in various asks / activities / processes and plan the implementation of energy efficient systems in a phased manner
- PC2.** segregate recyclable, non-recyclable and hazardous waste generated for disposal or efficient waste management

Respect diversity and strengthen practices to promote equity (equality)/inclusivity

To be competent, the user/individual on the job must be able to:

- PC3.** understand the diversity policy of the organization and use internal & external communication to colleagues to improve
- PC4.** comply with PwD inclusive policies for an adaptable and equitable work environment
- PC5.** improve through specifically designed recruitment practices, PwD friendly infrastructure, job roles, etc.
- PC6.** use and advocate for appropriate verbal/nonverbal communication, schemes and benefits of PwD.

Knowledge and Understanding (KU)

The individual on the job needs to know and understand:

- KU1.** the organization's policies and procedures about gender inclusivity, equality and sustainability while working with colleagues and your role and responsibilities in relation to this
- KU2.** inclusive tools and practices of communication to acknowledge/validate, share and promote the cause of gender parity at workplace. For example - supporting women with mentorship programs, speaking out against discriminatory practices or harassment
- KU3.** the concept of gender, gender equality and gender discrimination, and all forms of gender discrimination, violence and inequality, including the current and historical causes of gender inequality in the workplace

- KU4.** how to maintain and provide a conducive work environment that is free from any harassment. facilities and amenities to PwD to perform and excel in their role
- KU5.** organization's redressal mechanisms (like the POSH committee) to address harassment and bias at the workplace, with awareness of prevalent legislations against bias and sexual harassment
- KU6.** initiatives towards efficient use of natural resources and energy, reduction and prevention of pollution and promoting waste avoidance and recycling measures in line with internationally disseminated technologies and practices
- KU7.** all about various energy options including renewable and non-renewable with their environmental impacts, health issues, usage, safety and energy security
- KU8.** implications that any non-compliance with electricity and energy may have on individuals and the organization
- KU9.** the organization's electricity first aid emergency procedures
- KU10.** how to monitor, measure and report performance of environmental conservation
- KU11.** different types of electricity accidents, safety and security and how and when to report these
- KU12.** how to use the electricity/energy safety, accident reporting, emergency procedures and the importance of these

Generic Skills (GS)

User/individual on the job needs to know how to:

- GS1.** read PwD instructions, guidelines, procedures, diversity policies/acts, rules and service level agreements
- GS2.** be aware of one's own gender identity and gender role and respectful of the gender performances of others
- GS3.** organize team building or sensitization workshops to address gender biases, stereotypes and potentially blind spots
- GS4.** clarify personal norms and values related to energy production and usage as well as to reflect and evaluate their own energy usage in terms of efficiency and sufficiency
- GS5.** listen and communicate (oral) effectively and accurately on all PwD policies
- GS6.** apply balanced judgments in gender diversity situations
- GS7.** take action to reduce the carbon footprint of business activities and embed environmental responsibility
- GS8.** calibration session with employees to discuss gender biases, stereotypes and potentially blind spots

Assessment Criteria

Assessment Criteria for Outcomes	Theory Marks	Practical Marks	Project Marks	Viva Marks
<i>Sustainable Practices</i>	10	30	-	-
PC1. optimize usage of electricity/energy, materials, and water in various asks / activities / processes and plan the implementation of energy efficient systems in a phased manner	5	15	-	-
PC2. segregate recyclable, non-recyclable and hazardous waste generated for disposal or efficient waste management	5	15	-	-
<i>Respect diversity and strengthen practices to promote equity (equality)/inclusivity</i>	10	50	-	-
PC3. understand the diversity policy of the organization and use internal & external communication to colleagues to improve	5	10	-	-
PC4. comply with PwD inclusive policies for an adaptable and equitable work environment	-	10	-	-
PC5. improve through specifically designed recruitment practices, PwD friendly infrastructure, job roles, etc.	-	20	-	-
PC6. use and advocate for appropriate verbal/nonverbal communication, schemes and benefits of PwD.	5	10	-	-
NOS Total	20	80	-	-

National Occupational Standards (NOS) Parameters

NOS Code	SSC/N9014
NOS Name	Maintain an inclusive, environmentally sustainable workplace
Sector	IT-ITes
Sub-Sector	IT Services, Business Process Management, Engineering R&D, Software Product Development, Future Skills
Occupation	Generic,
NSQF Level	5
Credits	TBD
Version	1.0
Last Reviewed Date	NA
Next Review Date	31/03/2025
NSQC Clearance Date	

Assessment Guidelines and Assessment Weightage

Assessment Guidelines

1. Criteria for assessment for each Qualification Pack will be created by the Sector Skill Council. Each Performance Criteria (PC) will be assigned marks proportional to its importance in NOS. SSC will also lay down proportion of marks for Theory and Skills Practical for each PC.
2. The assessment for the theory part will be based on knowledge bank of questions created by the SSC.
3. Assessment will be conducted for all compulsory NOS, and where applicable, on the selected elective/option NOS/set of NOS.
4. Individual assessment agencies will create unique question papers for theory part for each candidate at each examination/training center (as per assessment criteria below).
5. Individual assessment agencies will create unique evaluations for skill practical for every student at each examination/training center based on this criterion.
6. To pass a QP, a trainee should score an average of 70% across generic NOS' and a minimum of 70% for each technical NOS
7. In case of unsuccessful completion, the trainee may seek reassessment on the Qualification Pack.

Minimum Aggregate Passing % at QP Level : 70

(**Please note:** Every Trainee should score a minimum aggregate passing percentage as specified above, to successfully clear the Qualification Pack assessment.)

Assessment Weightage

Compulsory NOS

National Occupational Standards	Theory Marks	Practical Marks	Project Marks	Viva Marks	Total Marks	Weightage
SSC/N8121.Evaluate technical performance of algorithmic models	30	70	-	-	100	25
SSC/N8122.Develop software code to support the deployment of algorithmic models	25	75	-	-	100	25
SSC/N9001.Manage your work to meet requirements	25	75	-	-	100	10
SSC/N9002.Work effectively with colleagues	20	80	-	-	100	10
SSC/N9004.Provide data/information in standard formats	25	75	-	-	100	10
SSC/N9006.Build and Maintain relationships in a Workplace	30	70	-	-	100	10
SSC/N9014.Maintain an inclusive, environmentally sustainable workplace	20	80	-	-	100	10
Total	175	525	-	-	700	100

Acronyms

NOS	National Occupational Standard(s)
NSQF	National Skills Qualifications Framework
QP	Qualifications Pack
TVET	Technical and Vocational Education and Training
IT-ITes	Information Technology - Information Technology enabled Services
BPM	Business Process Management
BPO	Business Process Outsourcing
KPO	Knowledge Process Outsourcing
LPO	LPO
IPO	Information Process Outsourcing
BCA	Bachelor of Computer Applications
B.Sc.	Bachelor of Science
OS	Occupational Standard(s)
NOS	National Occupational Standard(s)
QP	Qualifications Pack
UGC	University Grants Commission
MHRD	Ministry of Human Resource Development
MoLE	Ministry of Labour and Employment
NVEQF	National Vocational Education Qualifications Framework
NVQF	National Vocational Qualifications Framework
IT-ITes	Information Technology - Information Technology enabled Services
BPM	Business Process Management
BPO	Business Process Outsourcing
KPO	Knowledge Process Outsourcing
LPO	LPO



IPO	Information Process Outsourcing
BCA	Bachelor of Computer Applications
B.Sc.	Bachelor of Science
OS	Occupational Standard(s)
NOS	National Occupational Standard(s)
QP	Qualifications Pack
UGC	University Grants Commission
MHRD	Ministry of Human Resource Development
MoLE	Ministry of Labour and Employment
NVEQF	National Vocational Education Qualifications Framework
NVQF	National Vocational Qualifications Framework

Glossary

Sector	Sector is a conglomeration of different business operations having similar business and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Occupation	Occupation is a set of job roles, which perform similar/ related set of functions in an industry.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.
Occupational Standards (OS)	OS specify the standards of performance an individual must achieve when carrying out a function in the workplace, together with the Knowledge and Understanding (KU) they need to meet that standard consistently. Occupational Standards are applicable both in the Indian and global contexts.
Performance Criteria (PC)	Performance Criteria (PC) are statements that together specify the standard of performance required when carrying out a task.
National Occupational Standards (NOS)	NOS are occupational standards which apply uniquely in the Indian context.
Qualifications Pack (QP)	QP comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A QP is assigned a unique qualifications pack code.
Unit Code	Unit code is a unique identifier for an Occupational Standard, which is denoted by an 'N'
Unit Title	Unit title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is a set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on quality of performance required.
Knowledge and Understanding (KU)	Knowledge and Understanding (KU) are statements which together specify the technical, generic, professional and organisational specific knowledge that an individual needs in order to perform to the required standard.

Organisational Context	Organisational context includes the way the organisation is structured and how it operates, including the extent of operative knowledge managers have of their relevant areas of responsibility.
Technical Knowledge	Technical knowledge is the specific knowledge needed to accomplish specific designated responsibilities.
Core Skills/ Generic Skills (GS)	Core skills or Generic Skills (GS) are a group of skills that are the key to learning and working in today's world. These skills are typically needed in any work environment in today's world. These skills are typically needed in any work environment. In the context of the OS, these include communication related skills that are applicable to most job roles.
Electives	Electives are NOS/set of NOS that are identified by the sector as contributive to specialization in a job role. There may be multiple electives within a QP for each specialized job role. Trainees must select at least one elective for the successful completion of a QP with Electives.
Options	Options are NOS/set of NOS that are identified by the sector as additional skills. There may be multiple options within a QP. It is not mandatory to select any of the options to complete a QP with Options.
Sector	Sector is a conglomeration of different business operations having similar businesses and interests. It may also be defined as a distinct subset of the economy whose components share similar characteristics and interests.
Sub-sector	Sub-sector is derived from a further breakdown based on the characteristics and interests of its components.
Vertical	Vertical may exist within a sub-sector representing different domain areas or the client industries served by the industry.
Occupation	Occupation is a set of job roles, which perform similar/related set of functions in an industry.
Function	Function is an activity necessary for achieving the key purpose of the sector, occupation, or area of work, which can be carried out by a person or a group of persons. Functions are identified through functional analysis and form the basis of OS.
Sub-functions	Sub-functions are sub-activities essential to fulfill the achieving the objectives of the function.
Job role	Job role defines a unique set of functions that together form a unique employment opportunity in an organisation.

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Performance Criteria	Performance Criteria are statements that together specify the standard of performance required when carrying out a task.
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Qualifications Pack Code	Qualifications Pack Code is a unique reference code that identifies a qualifications pack.
Qualifications Pack(QP)	Qualifications Pack comprises the set of OS, together with the educational, training and other criteria required to perform a job role. A Qualifications Pack is assigned a unique qualification pack code.
Unit Code	Unit Code is a unique identifier for an OS unit, which can be denoted with either an O or an N.
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Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
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Knowledge and Understanding	Knowledge and Understanding are statements which together specify the technical, generic, professional and organizational specific knowledge that an individual needs in order to perform to the required standard.
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Helpdesk	Helpdesk is an entity to which the customers will report their IT problems. IT Service Helpdesk Attendant is responsible for managing the helpdesk.
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Unit Title	Unit Title gives a clear overall statement about what the incumbent should be able to do.
Description	Description gives a short summary of the unit content. This would be helpful to anyone searching on a database to verify that this is the appropriate OS they are looking for.
Scope	Scope is the set of statements specifying the range of variables that an individual may have to deal with in carrying out the function which have a critical impact on the quality of performance required.
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